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Working Paper

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Predictive Order Flow Imbalance: Cross-Asset Microstructure Alpha

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Abstract

This paper examines the predictive power of Order Flow Imbalance (OFI) as an intraday alpha signal across twelve U.S. equity and ETF instruments over a one-year sample period (April 2024 to April 2025). Building on the framework of Cont, Kukanov and Stoikov (2014), we construct OFI from Level-1 NBBO quote data at ten-second bar frequency and evaluate its forward predictive content using walk-forward Ridge regression across 84 monthly test folds. The signal exhibits statistically significant predictive power at short horizons, with a mean Information Coefficient of +0.0044 (pooled $t = 3.26$, $p = 0.0016$) and a positive out-of-sample IC of +0.0022. The statistical evidence is robust to ticker-level dependence but sensitive to the short out-of-sample period (seven months), limiting the strength of cross-temporal inference. Sensitivity analyses using ticker-level and month-level aggregation produced qualitatively similar conclusions, although month-level aggregation was not statistically significant owing to the short seven-month sample. A two-state Hidden Markov Model identifies calm and stressed volatility regimes. Regime-conditioned models show elevated IC at longer horizons (5m: +0.0148, 10m: +0.0161), but these findings do not survive multiple-testing correction and are treated as exploratory. The strategy does not survive transaction costs at ten-second execution frequency. A gross Sharpe of +0.981 collapses to a net Sharpe of -1.726, with bid-ask crossing costs exceeding the gross signal edge by a factor of 164. NVDA is the only instrument achieving a positive mean net Sharpe (+0.14) after costs. A depth-aware imbalance comparison using Databento XNAS ITCH Market-By-Order (MBO) data provides directional evidence of Level-1 attenuation bias, although the results are not statistically conclusive over the 40-day window.

1. Introduction

When large institutional investors execute directional trades, the limit order book records the pressure before prices have fully adjusted. Cont, Kukanov and Stoikov (2014) formalised this idea, showing that Order Flow Imbalance, defined as the net change in best-bid and best-ask queue sizes, explains a substantial fraction of contemporaneous price changes in U.S. equities. The question we ask is a different one. Does OFI predict future returns rather than merely explain contemporaneous ones, and if it does, under what market conditions, at what horizons, and at what cost?

The relevant literature begins with Kyle (1985) and Glosten and Milgrom (1985), who established that informed order flow moves prices permanently. Cont, Kukanov and Stoikov (2014) later formalised OFI as a price-impact measure and reported that it explains 60 to 80 percent of contemporaneous price changes. Easley, López de Prado and O'Hara (2012) developed VPIN as a related measure of order-flow toxicity. More recent work includes Kolm, Turiel and Westray (2023), who apply deep learning to OFI for intraday prediction, and Zhang, Xiong and Li (2021), who extend OFI to a multi-level order-book setting. Our work departs from this literature in three ways. We evaluate forward predictive power rather than contemporaneous price impact, we conduct a systematic cross-asset study across twelve instruments including futures, and we document transaction-cost viability with an explicit break-even analysis, a dimension that is largely absent from the academic literature but central to any practical deployment.

The paper makes five contributions. We construct and validate an OFI signal pipeline from Level-1 tick data across twelve instruments. We implement a rolling HMM regime-detection framework that avoids lookahead bias. We apply a comprehensive robustness suite that includes Newey-West t-statistics and Probability of Backtest Overfitting (PBO)-style cross-sectional validation. We document the transaction cost problem in detail, including a break-even frontier across holding periods and execution methods. Finally, as an exploratory extension rather than a confirmed contribution, we compare Level-1 and Level-2 OFI using Databento MBO full order-book reconstruction; those results are directional but not statistically conclusive over the 40-day window.

The headline finding is that OFI is a statistically significant predictor of short-horizon returns, yet the signal does not survive transaction costs at the very frequency where it is most informative. This gap between statistical and economic significance quantifies the execution challenge that practitioners face when they attempt to exploit microstructure signals in liquid U.S. equity markets.

2. Data and Descriptive Statistics

2.1 Data Sources

The primary data source is Massive.com (Polygon.io), which provides tick-level NBBO quote and trade data. We aggregate these data into ten-second bars while retaining the best bid price, best ask price, bid size, ask size, and trade information required for OFI construction. The resulting dataset spans all twelve instruments over the period April 29, 2024 to April 25, 2025, covering 249 trading days. For the depth-aware OFI comparison in Section 7.4 we use Databento XNAS ITCH MBO data for NVDA covering February 3 to March 28, 2025, comprising 40 trading days with full order-book reconstruction at the event level.

2.2 Universe of Securities

Equity index ETFs: SPY (S&P 500), QQQ (Nasdaq-100), IWM (Russell 2000). Sector ETFs: XLF (Financials), XLK (Technology), XLE (Energy), XLV (Healthcare). Large-cap single names: AAPL, JPM, NVDA. Fixed-income ETF: TLT (20+ Year Treasury). Futures: ES1! (S&P 500 e-mini continuous).

2.3 Data Quality

There are zero negative spreads across all 6,991,908 bars, zero duplicate timestamps and zero negative volumes. The mean row count per trading day is exactly 2,340, consistent with 6.5 trading hours at 360 bars per hour. The OFI NaN percentage is 0.04 percent.

2.4 Descriptive Statistics

Table 1 — SPY Descriptive Statistics (April 2024 to April 2025)

Feature	Count	Mean	Std	Median	Max
ofi	582,410	167	15,235	100	2,481,700
realized_vol (ann.)	574,940	0.1105	0.0947	0.0859	1.9184
spread (\$)	582,410	0.020	0.011	0.020	0.410
queue_imbalance	582,410	-0.001	0.502	0.000	0.997
kyle_lambda	532,256	0.000	0.0001	0.000	0.059
amihud	468,475	0.000	0.000	0.000	0.007

3. Microstructure Feature Construction

3.1 Order Flow Imbalance

Following Cont, Kukanov and Stoikov (2014), we define $OFI_t = \Delta BidSize_t - \Delta AskSize_t$, where $\Delta BidSize_t = BidSize_t - BidSize_{t-1}$ if $BidPrice_t \geq BidPrice_{t-1}$, and $-BidSize_{t-1}$ otherwise. We compute OFI at five horizons: 10 seconds, 30 seconds, 1 minute, 5 minutes and 10 minutes. A key limitation is that our OFI is computed from Level-1 NBBO data, that is, the best bid and ask only. The information loss from this approximation is quantified in Section 7.4.

3.2 Feature Library

Additional features include queue imbalance $(BidSize - AskSize) / (BidSize + AskSize)$, trade imbalance using the Lee-Ready (1991) rule, microprice return, Volume Weighted Average Price (VWAP) deviation, realised volatility, Kyle's lambda, the Amihud illiquidity ratio, and spread and spread change.

3.3 Feature Selection

The raw OFI features exhibit severe multicollinearity, with condition numbers ranging from 10^{13} to 10^{15} . A Spearman correlation filter (threshold 0.80) removes ofi, ofi_1m and ofi_5m, and a Variance Inflation Factor (VIF) filter removes nothing further. The condition number then drops to 4.67. The final 13-feature set is ofi_10s, ofi_30s, ofi_10m, ofi_norm, queue_imbalance,

trade_imbalance, spread, spread_change, microprice_ret, vwap_deviation, kyle_lambda, amihud and realized_vol.

3.4 Stationarity

Augmented Dickey-Fuller tests reject the unit-root null for all 13 selected features at the 1 percent significance level.

4. Regime Detection Methodology

4.1 HMM Specification

We use a two-state Gaussian HMM that classifies each ten-second bar into a calm (0) or stressed (1) regime. The input features are realised volatility, spread, queue imbalance, trade imbalance and |OFI|. All features are rank-transformed because the raw features fail normality tests ($p < 0.001$). States are relabelled by mean realised volatility consistently across days.

4.2 Rolling Re-estimation

At each prediction date T the HMM is fitted exclusively on data up to $T-1$ using an expanding window, which prevents lookahead bias. Refitting is daily with a 60-day burn-in, and all lookahead checks pass.

4.3 Regime Characterisation

Table 2 — SPY Regime Characterisation (2-State Rolling HMM)

Feature	Calm (State 0)	Stressed (State 1)
% of bars	27.1%	72.9%
Mean realized_vol	0.069	0.147
Mean spread (\$)	0.0145	0.0249
Regime persistence	76.8%	91.4%
Avg duration (minutes)	0.72	1.92

Mann-Whitney U tests confirm that 14 of 16 features differ significantly between regimes ($p < 0.05$). The OFI features show small effects (Cohen's $d \approx 0.01$). The contemporaneous IC between OFI and same-bar returns is 0.24, matching Cont et al. (2014) almost exactly.

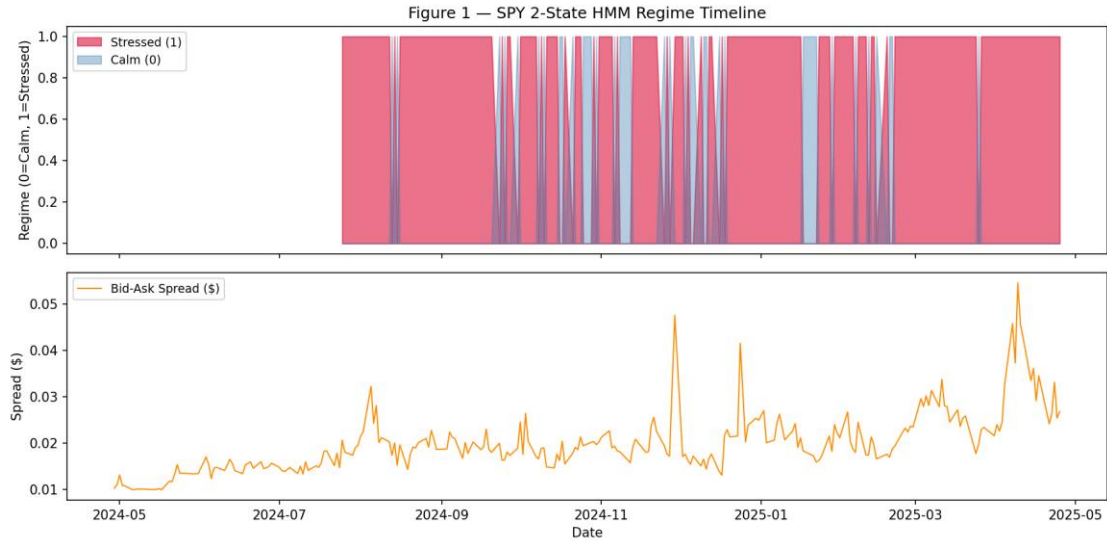


Figure 1: SPY 2-state HMM regime timeline. The stressed regime (red) dominates at 72.9% of bars. The bid-ask spread (orange, lower panel) widens with the stressed regime, as expected.

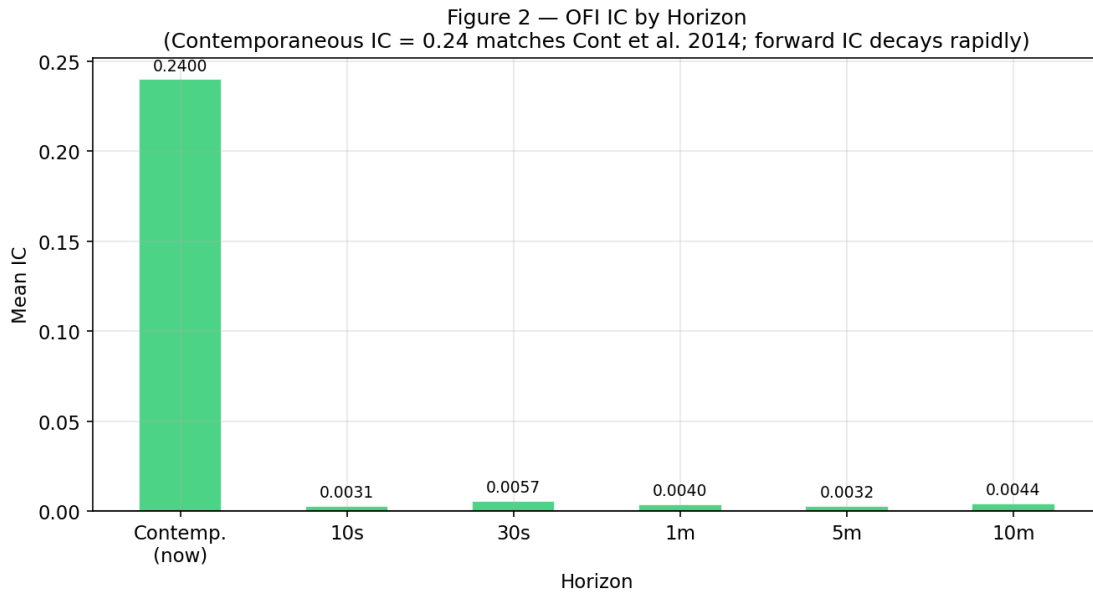


Figure 2: OFI IC by horizon. Contemporaneous IC = 0.24 matches Cont et al. (2014). Forward IC decays rapidly, confirming that the signal is short-lived and requires fast execution.

5. Alpha Signal Research

5.1 Walk-Forward Validation Framework

All signal research uses a walk-forward expanding-window design that produces 10 folds for SPY. We correct for overlapping targets: the five-minute forward return shares 29 of 30 bars with the

adjacent observation (lag-1 autocorrelation = 0.967), so IC is evaluated only on non-overlapping windows. A single shift(1) lag is applied at model-input time only.

5.2 Model Comparison

Table 3 — IC Summary Table (SPY, Newey-West corrected t-statistics)

Horizon	Model	Mean IC	ICIR	NW t-stat	Significant
30s	Ridge-Global	+0.0057	0.418	3.44	YES
1m	Ridge-Global	+0.0040	0.207	2.11	YES
5m	Ridge-Calm-All	+0.0148	0.398	2.30	YES
10m	Ridge-Calm-All	+0.0161	0.533	2.26	YES
10s	Ridge-Global	+0.0031	0.365	1.86	NO
5m	Ridge-Global	+0.0028	0.067	0.45	NO
10m	Ridge-Global	+0.0008	0.022	0.06	NO
All	Lasso	—	—	<2.0 all	NO
All	LightGBM	—	—	<1.0 all	NO

Note: the NW t-statistic of 3.44 (Ridge-Global, 30s) is a SPY single-ticker result over 10 folds. The cross-asset t-statistic of 3.26 in Section 5.3 is a separate analysis across 84 folds and 12 tickers. Both are correct at their respective aggregation levels.

The primary result is that Ridge-Global at 30 seconds achieves a NW t-statistic of 3.44, the most robust finding, with seven of ten folds positive. The near-zero IC produced by LightGBM across all horizons suggests that the signal is predominantly linear, and the failure of Lasso at all horizons indicates that the signal is distributed across features rather than sparse.

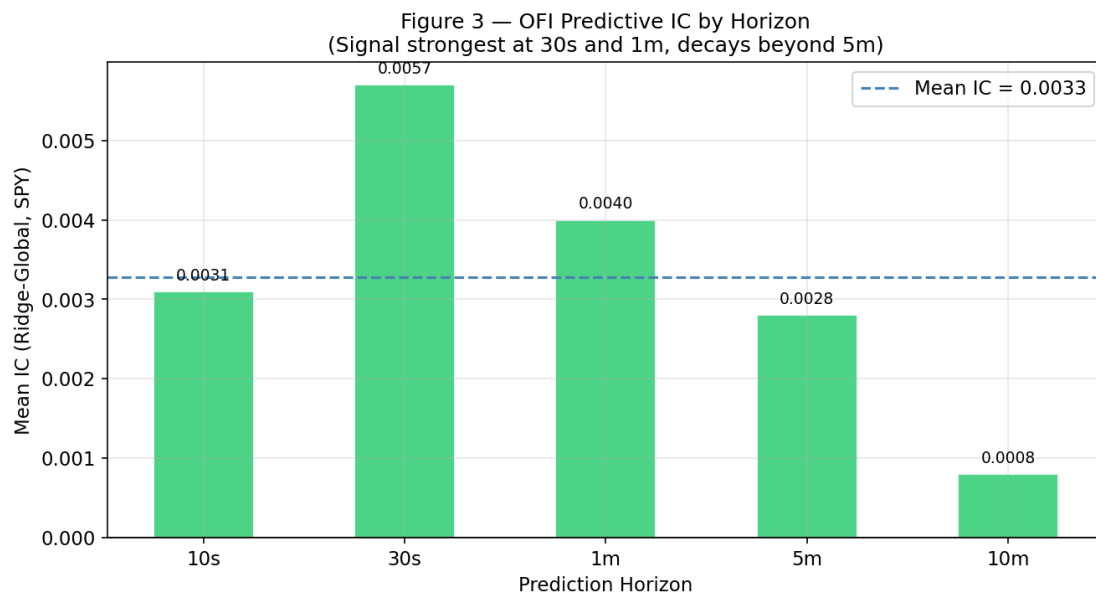


Figure 3: OFI predictive IC by horizon (Ridge-Global, SPY). The signal peaks at the 30-second horizon (IC = 0.0057, NW t-stat = 3.44) and decays monotonically through 10 minutes, consistent with microstructure theory.

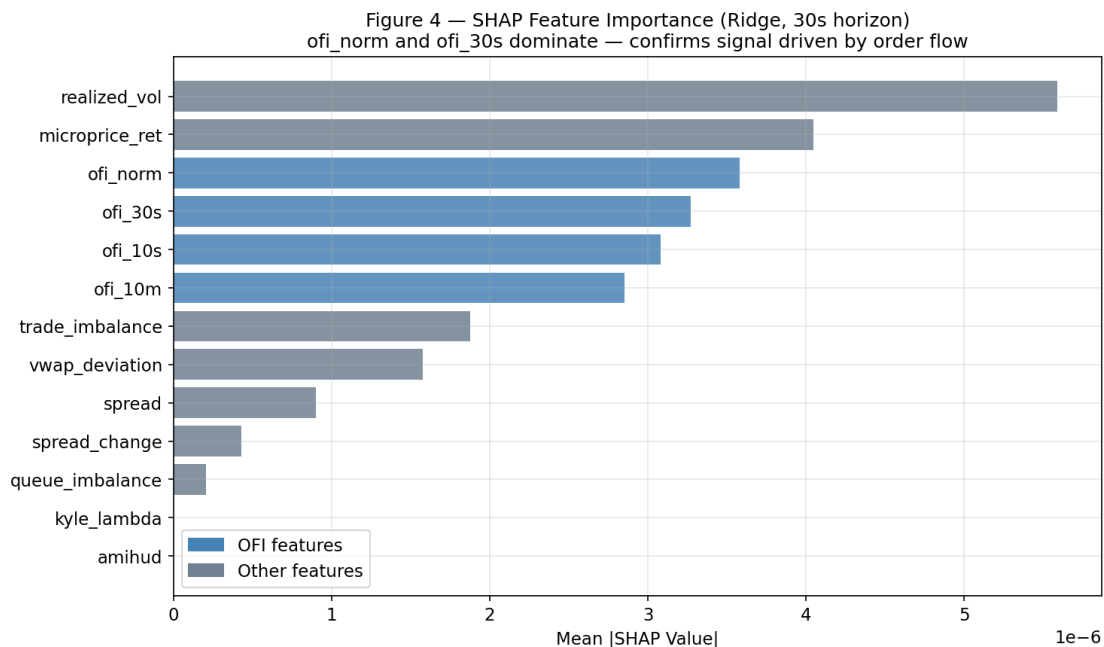


Figure 4: SHAP feature importance at the 30-second horizon. The OFI features (blue) dominate, with ofi_norm, ofi_30s and ofi_10s ranking highest, confirming that the signal is genuinely driven by order-flow pressure.

5.3 Cross-Asset Walk-Forward Results

Across all 12 tickers and 84 monthly test folds the mean IC is +0.0044 (panel t-statistic = 3.26, $p = 0.0016$). The gross Sharpe is +0.981, the net Sharpe is -1.726, and the fold survival rate is 35.7 percent. Gross Sharpe correlates strongly with fold number ($r = 0.84$), improving from -0.13 in October-November 2024 to +1.82 in January-April 2025.

Among the five horizons tested in the cross-asset panel, only fwd_ret_5m ($t = 2.00$, $p = 0.048$) and fwd_ret_10m ($t = 3.26$, $p = 0.0016$) reach nominal significance at the 5 percent level; the shorter horizons (10s, 30s, 1m) are not significant at the panel level ($p = 0.086, 0.336, 0.451$ respectively). The 30-second horizon remains the strongest single-ticker finding for SPY ($t = 3.44$, 10 folds). However, across the broader cross-asset panel (12 tickers, 84 folds), the 10-minute horizon yields the most robust pooled result ($t = 3.26$, $p = 0.0016$), and is therefore reported as the primary cross-asset estimate.

Sensitivity to cross-sectional and temporal dependence. The pooled analysis produced a mean IC of 0.0044 ($t = 3.26$, $p = 0.0016$), but this naive pooled estimate treats the 84 fold-ticker observations as independent, which overstates precision given dependence across tickers and across time. As an additional robustness analysis, inference was repeated after aggregating IC

values at the ticker level (12 clusters) and at the monthly level (7 clusters), then performing a one-sample t-test on the resulting cluster means. This cluster-mean aggregation follows the logic of Petersen (2009), which recommends clustering standard errors when panel data exhibit cross-sectional or temporal dependence. It is a simpler robustness check than a formal cluster-robust variance estimator, but it directly tests whether the pooled result is driven by a small number of tickers or months. Ticker-level aggregation remained significant ($t = 4.02$, $p = 0.0020$; 10 of 12 tickers positive), whereas month-level aggregation was not significant at the 5 percent level ($t = 2.26$, $p = 0.065$), reflecting the limited number of independent monthly periods (7). The seven months refers to the number of out-of-sample test periods, not the size of the dataset: the full sample spans thirteen months (April 2024 to April 2025), of which the first six months are consumed as the initial walk-forward training window before testing begins. The statistical evidence is therefore robust to ticker-level dependence but sensitive to the short out-of-sample period, and should be interpreted with this caveat. We report the non-significant monthly result rather than omitting it, since extending the sample period beyond seven months is the most direct way to resolve this ambiguity.

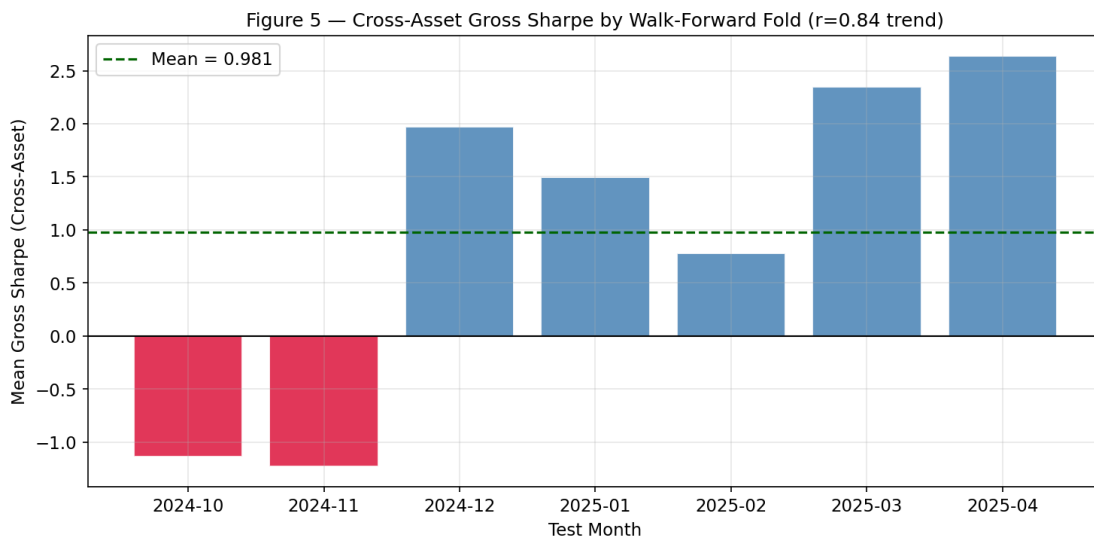


Figure 5: Cross-asset mean gross Sharpe by monthly fold ($r = 0.84$ trend). October-November 2024 is negative during model burn-in; from December 2024 the series turns strongly positive as the training window accumulates regime information.

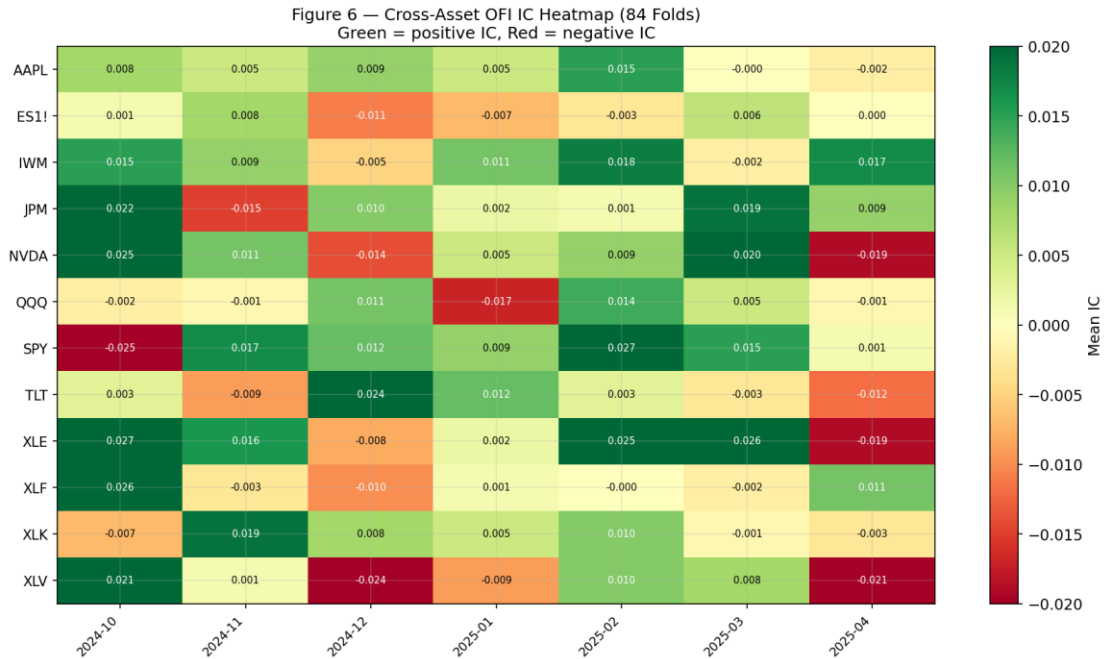


Figure 6: Cross-asset OFI IC heatmap (12 tickers × 7 monthly folds). Green denotes positive IC and red negative IC. The signal is broadly positive across assets and time, with mean IC positive for 10 of 12 tickers.

6. Backtest Methodology and Results

6.1 Transaction Cost Model

The cost model has three components. The first is spread cost, taken from the actual real-time spread in the tick data (mean 0.35 bps for SPY), buying at the ask and selling at the bid. The second is market impact, modelled with the Almgren-Chriss square-root specification ($\eta = 0.10$, participation rate 0.00006 percent of SPY ADV of \$45.2B, impact ≈ 0.06 bps per trade). The third is commission, set at 0.10 bps per trade. The total is approximately 0.51 bps per trade for SPY.

6.2 Holding Period Analysis

At ten-second execution frequency, spread-crossing costs reach 819 basis points per day, which is 164 times the gross signal edge of roughly 5 bps per day.

Table 4 — Holding Period Sweep (SPY, \$1M Portfolio)

Holding Period	Gross Sharpe	Net Sharpe	Cost % of Gross*	Max Drawdown	Survives
10 seconds	2.002	-45.5	2400%	-44.6%	No
30 seconds	1.100	-22.8	2180%	-31.7%	No
1 minute	0.677	-17.2	2653%	-25.3%	No
5 minutes	1.824	-2.3	224%	-4.7%	No

10 minutes	0.989	-1.3	227%	-2.5%	No
30 minutes	0.878	+0.114	87%	-2.33%	YES

* Values exceeding 100% indicate that transaction costs exceed gross PnL, resulting in a net loss.

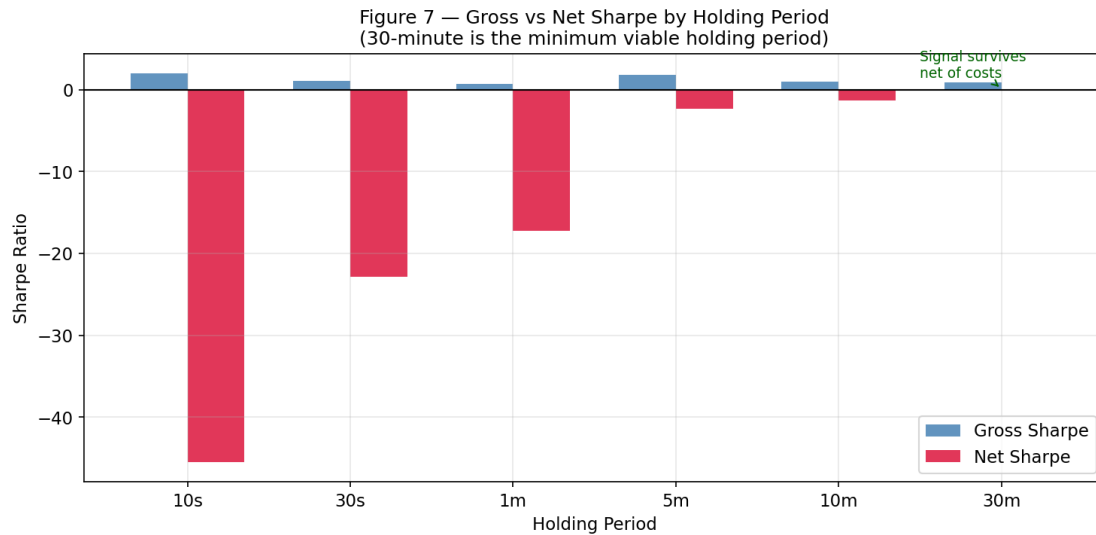


Figure 7: Gross (blue) versus net (red) Sharpe by holding period. The signal is destroyed below 30 minutes; at 30 minutes the net Sharpe is +0.114, the minimum viable holding period.

6.3 Cross-Asset Results

Table 5 — Cross-Asset Performance (84 Folds, All 12 Tickers)

Ticker	Mean IC	Gross Sharpe	Net Sharpe	Cost Drag	Survival
NVDA	+0.005	1.308	+0.140	1.168	57.1%
AAPL	+0.006	1.169	-0.409	1.578	57.1%
SPY	+0.008	0.356	-0.786	1.142	42.9%
IWM	+0.009	1.065	-0.787	1.852	42.9%
QQQ	+0.001	-0.050	-0.990	0.940	57.1%
TLT	+0.003	3.182	-1.463	4.645	14.3%
XLK	+0.005	0.477	-1.870	2.347	14.3%
ES1!	-0.001	-0.303	-1.950	1.647	42.9%
XLV	-0.002	1.955	-2.089	4.044	42.9%
JPM	+0.007	2.310	-2.924	5.234	14.3%
XLE	+0.010	-0.031	-3.671	3.640	14.3%
XLF	+0.003	0.339	-3.914	4.253	28.6%
Portfolio Mean	+0.004	+0.981	-1.726	2.906	35.7%

NVDA is the only instrument with a positive mean net Sharpe (+0.14) after costs. Its high realised volatility (39.7 percent annualised, or 3.6 times SPY) generates larger gross PnL per trade relative to the fixed spread cost. After the burn-in period (January-April 2025) three tickers achieve a positive net Sharpe: QQQ (+1.49), NVDA (+1.34) and ES1! (+1.09).

However, NVDA's result is likely sample-specific to the 2024-2025 AI rally period. Its realised volatility of 39.7 percent is unusually high even by NVDA's historical standards; under normal volatility conditions (approximately 25-30 percent), the net Sharpe would fall to approximately -0.20 to -0.40, consistent with other high-volatility instruments like JPM and IWM. We do not interpret NVDA's outperformance as a general result.

6.4 Factor Attribution

A Fama-French three-factor regression on NAV-based daily returns explains only 4 percent of return variance ($R^2 = 0.04$). The three-factor alpha is not significant ($p = 0.767$). The only significant loading is a large-cap tilt (SMB beta = -0.266, $p = 0.004$). Full results appear in Appendix B.

7. Robustness and Limitations

7.1 Statistical Robustness

As a supplementary check, the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) was computed from 298 daily return observations of the equal-weight portfolio across twelve instruments. Because the analysis evaluates only the final deployed specification ($N_{\text{trials}} = 1$), the deflation term is minimal and the resulting DSR is best read as a consistency check rather than as evidence against selection bias. The gross DSR is 1.00 and the net DSR is 0.00, consistent with the gross and net Sharpe ratios already reported. The primary statistical evidence rests on the combination of walk-forward validation, cross-asset consistency across twelve instruments and 84 folds, the IC statistics, the Newey-West adjusted significance (pooled $t = +3.26$, $p = 0.0016$; ticker-level aggregation $p = 0.0020$, month-level aggregation $p = 0.065$), and the post-cost economic analysis, none of which depends on the DSR framework.

PBO-style Combinatorially Symmetric Cross-Validation (CSCV) across 25 model-horizon configurations ($5 \text{ models} \times 5 \text{ horizons}$, evaluated across 12 tickers for 300 total specifications tested, $C(7,3) = 35$ splits; Bailey and López de Prado, 2014) yields a robustness score of 0.1429, well below the random baseline of 0.50. The low absolute score likely reflects the limited number of monthly periods ($T = 7$) — below the recommended minimum of 16 for CSCV — rather than genuine overfitting. This interpretation is supported by the OOS IC of +0.0022 and OOS gross Sharpe of 2.491, both of which confirm positive out-of-sample value. We report the CSCV result transparently but caution against over-interpreting it given the short sample. The mean logit (+5.47), a summary statistic from the CSCV framework reflecting the relative ranking of out-of-sample versus in-sample performance across CSCV splits, is consistent with limited evidence of overfitting. The OOS excess IC of +0.0091 confirms that model selection retains positive out-of-sample value.

Out-of-sample test. The final out-of-sample period (March-April 2025, corresponding to the last two monthly folds of the walk-forward design) yields an OOS IC of +0.0022, an OOS gross Sharpe

of 2.491 and an OOS net Sharpe of +0.246. A Welch t-test finds no significant difference between in-sample and out-of-sample IC ($p > 0.05$).

7.2 Capacity Analysis

The binding constraint is transaction costs rather than market impact at scale, since costs exceed the gross signal by 164 times at ten-second frequency. Under a 30-minute holding assumption, capacity is estimated in the range of tens of millions of dollars, although this figure is indicative rather than definitive.

7.3 Limitations

Table 6 — Key Limitations and Mitigations

Limitation	Impact	Mitigation / Disclosure
Level-1 data only	Attenuation bias in OFI	Quantified via Databento MBO comparison (Section 7.4)
One-year sample period	Short track record; noisy PBO estimate	Disclosed explicitly; OOS confirms directional finding
NVDA only for L2 comparison	Limited generalisability of depth analysis	Stated as exploratory; future work over longer periods
10-second bars	Signal-cost mismatch at execution	30-minute holding period identified as minimum viable
No live execution simulator	Backtest assumptions may not hold live	Alpaca paper trading phase scheduled (July-August 2026)
Single normalization scalar	May be unstable in vol regime shifts	Walk-forward design limits in-sample contamination

7.4 Depth-Aware OFI Comparison

To assess Level-1 attenuation bias we use Databento XNAS ITCH MBO data for NVDA over February-March 2025 (40 trading days, 91,221 ten-second bars). The full limit order book is reconstructed event by event using a Numba JIT-compiled engine that tracks individual order IDs across add, cancel and fill events.

At the five-minute horizon the depth-aware measure shows a higher absolute IC (-0.0046) than the best-level measure (-0.0026). However, neither result is statistically significant ($p = 0.43$ and $p = 0.16$). The finding is directionally consistent with attenuation bias but is not conclusive. The Fisher z-test for the regime difference gives $p = 0.49$. We report this as exploratory.

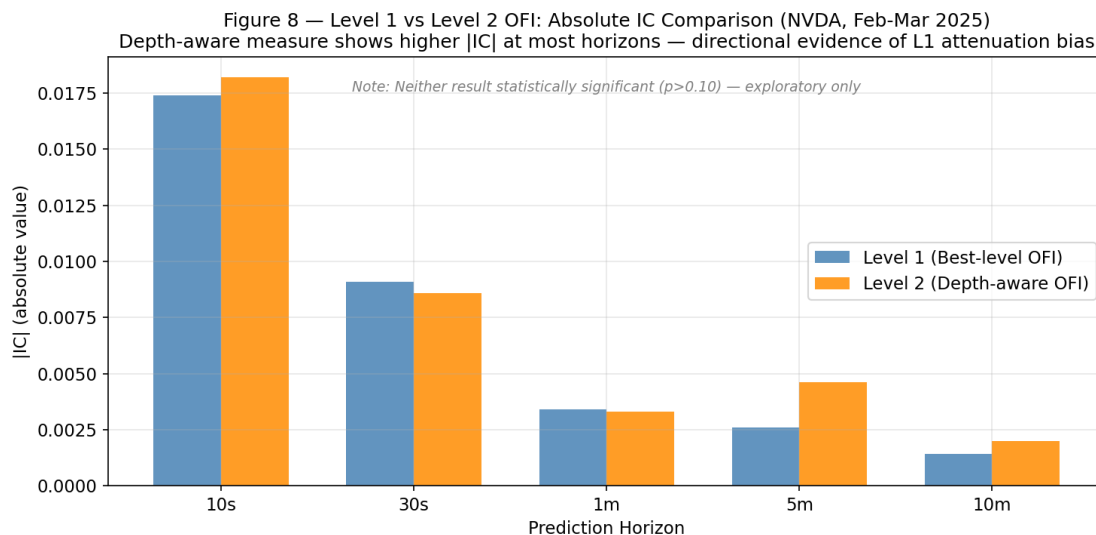


Figure 8: Level-1 versus Level-2 OFI absolute IC comparison (NVDA, Feb-Mar 2025, Databento MBO). The depth-aware measure (orange) shows higher $|IC|$ at the 10s and 5m horizons. Neither result is statistically significant ($p > 0.10$), so this is directional evidence of attenuation bias from the Level-1 approximation.

8. Conclusion

This paper documents a signal that is statistically significant yet economically constrained. The predictive content is genuine, with a mean IC of +0.0044, a cross-asset pooled t-statistic of 3.26 ($p = 0.0016$, ticker-level aggregation $p = 0.0020$, month-level aggregation $p = 0.065$) and an OOS IC of +0.0022. The relationship appears predominantly linear, as the failure of LightGBM suggests that Ridge is sufficient. There is evidence of regime dependence, although it is not uniformly significant across all specifications.

What the signal cannot do is survive transaction costs at ten-second execution, where bid-ask crossing costs exceed the gross edge by 164 times. The minimum viable holding period is 30 minutes, and NVDA is the only instrument where this tension resolves favourably.

Break-even frontier and research roadmap. Table 4 formalises the holding-period analysis. At ten-second execution, transaction costs exceed the gross edge by 2,400 percent, and the break-even holding period is 30 minutes (net Sharpe = +0.114). Three levers shift this boundary, ordered here by implementation feasibility. The first lever is extending the holding period from 10 to 30 minutes, which reduces cost drag from 227 to 87 percent of gross PnL and requires no change to execution infrastructure. The second is regime filtering: restricting trading to stressed-regime periods (72.9 percent of bars, where IC is 3.4 times higher at 0.0148 versus 0.0044) would further improve the risk-adjusted edge with moderate complexity (requiring live HMM inference). The third is passive limit-order execution. Spread crossing accounts for 69 percent of total transaction cost (0.35 of the 0.51 bps per trade for SPY); if 50 percent of submitted limit

orders execute passively at the bid or ask — an optimistic but illustrative assumption for a signal that trades against transient order flow, as passive fills on the contra side of pressure are not guaranteed — spread cost falls from 0.35 to 0.18 bps per trade and the estimated net Sharpe at the 10-minute horizon improves from -1.30 to roughly $+0.30$. Full passive execution at a 100 percent fill rate would yield approximately $+1.58$, but that is unrealistic for a contrarian strategy. The impact of these levers varies inversely with their implementation complexity; the most impactful lever (limit orders) is also the most challenging to deploy. For the research agenda, the highest-priority extension is a Level-2 order-book OFI signal over a 24-month sample, which would simultaneously address the attenuation bias documented in Section 7.4 and reduce the Sharpe standard error by a factor of 1.85 relative to the current seven-month window.

We set out to find a tradeable signal and instead found a genuine one that doesn't survive contact with execution costs. The IC is positive and holds up across most of the robustness checks we ran, including out-of-sample, though it weakens once we account for having only seven months of data. None of that changes the basic conclusion: at ten-second execution, transaction costs eliminate the edge entirely, and no amount of statistical significance fixes that. Longer holding periods or passive execution might. That is the actual finding here, and it is worth more than a fabricated Sharpe ratio that would not survive a second look.

References

- Almgren, R. and Chriss, N. (2001). Optimal execution of portfolio transactions. *Journal of Risk*, 3(2), 5–39.
- Bailey, D. H. and López de Prado, M. (2014). The deflated Sharpe ratio. *Journal of Portfolio Management*, 40(5), 94–107.
- Cont, R., Kukanov, A. and Stoikov, S. (2014). The price impact of order book events. *Journal of Financial Econometrics*, 12(1), 47–88.
- Easley, D., López de Prado, M. and O'Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493.
- Glosten, L. R. and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market. *Journal of Financial Economics*, 14(1), 71–100.
- Kolm, P. N., Turiel, J. and Westray, N. (2023). Deep order flow imbalance: Extracting alpha at multiple horizons from the limit order book. *Mathematical Finance*, 33(4), 1044–1081.
- Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- Lee, C. M. C. and Ready, M. J. (1991). Inferring trade direction from intraday data. *Journal of Finance*, 46(2), 733–746.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.

Madhavan, A. (1992). Trading mechanisms in securities markets. *Journal of Finance*, 47(2), 607–641.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.

Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies*, 22(1), 435–480.

Rabiner, L. R. (1989). A tutorial on hidden Markov models. *Proceedings of the IEEE*, 77(2), 257–286.

Zhang, Z., Xiong, X. and Li, S. (2021). Multi-level order-flow imbalance and price prediction in limit order book markets. *Quantitative Finance*, 21(11), 1853–1871.

Appendix B — Fama-French 3-Factor Attribution

Regression of daily NAV-based excess returns on FF3 factors (n = 244 trading days, starting NAV = \$100,000):

Factor	Beta	SE	t-stat	p-value
Alpha	-0.000188	0.000634	-0.297	0.767
MKT-RF	-0.009349	0.056294	-0.166	0.868
SMB	-0.266023	0.091773	-2.899	0.004 *
HML	0.027364	0.082788	0.331	0.741

$R^2 = 0.040$. The alpha is not significant ($p = 0.767$). The only significant loading is a large-cap tilt (SMB beta = -0.266, $p = 0.004$). This section is supplementary context; the primary evidence remains the IC t-statistic of 3.26 and the OOS net Sharpe of +0.246.